

Question 1.1: Building the Engine

3-Layer FCNN from Scratch (NumPy Only)

Dataset: UCI Adult Census Income

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1 Introduction

This report documents the implementation of a 3-layer Fully Connected Neural Network using only NumPy. The goal is to understand neural network fundamentals by manually coding forward propagation, backpropagation, and gradient descent without any deep learning frameworks.

2 Network Architecture

The implemented network consists of:

- **Input Layer:** 108 features
- **Hidden Layer 1:** 64 neurons, ReLU activation
- **Hidden Layer 2:** 32 neurons, ReLU activation
- **Output Layer:** 1 neuron, Sigmoid activation

Training Configuration:

- Weight Initialization: He initialization
- Optimizer: Vanilla SGD
- Learning Rate: 0.01
- Loss Function: Binary Cross-Entropy
- Iterations: 1,000

3 Implementation Components

3.1 Layer Initialization

He initialization was used for weights:

```
```python
```

```
w1 = np.random.randn(64, 108) * np.sqrt(2/108)
```

```
b1 = np.zeros((64, 1))
```

```
...
```

### ### 3.2 Forward Pass

```
```python
z1 = np.dot(w1, X) + b1
a1 = relu(z1)
z2 = np.dot(w2, a1) + b2
a2 = relu(z2)
z3 = np.dot(w3, a2) + b3
a3 = sigmoid(z3)
```
```

### ### 3.3 Backpropagation

Gradients computed using chain rule:

```
```python
dz3 = a3 - Y
dw3 = np.dot(dz3, a2.T) / m
# ... similar for other layers
```
```

### ### 3.4 Weight Updates

```
```python
w1 -= lr * dw1
b1 -= lr * db1
# ... similar for other layers
```
```

```

```

## ## 4 Training Results

| Step  | Loss   |
|-------|--------|
| ----- | -----  |
| 0     | 1.1784 |
| 100   | 0.5625 |
| 200   | 0.5567 |

| 300 | 0.5517 |  
| 400 | 0.5456 |  
| 500 | 0.5428 |  
| 600 | 0.5407 |  
| 700 | 0.5390 |  
| 800 | 0.5375 |  
| 900 | 0.5363 |

**\*\*Final Accuracy:\*\* 76.39%**

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## ## 5 Observations

### ### 5.1 Convergence

- Rapid decrease in first 100 iterations
- Steady improvement throughout training
- No instability or divergence

### ### 5.2 Performance

- Exceeds 75% requirement
- Demonstrates successful from-scratch implementation

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## ## 6 Conclusion

The from-scratch implementation successfully:

1. Achieved 76.39% accuracy (exceeds 75% requirement)
2. Demonstrated stable training without frameworks
3. Validated correct backpropagation implementation
4. Provided deep understanding of neural network mechanics